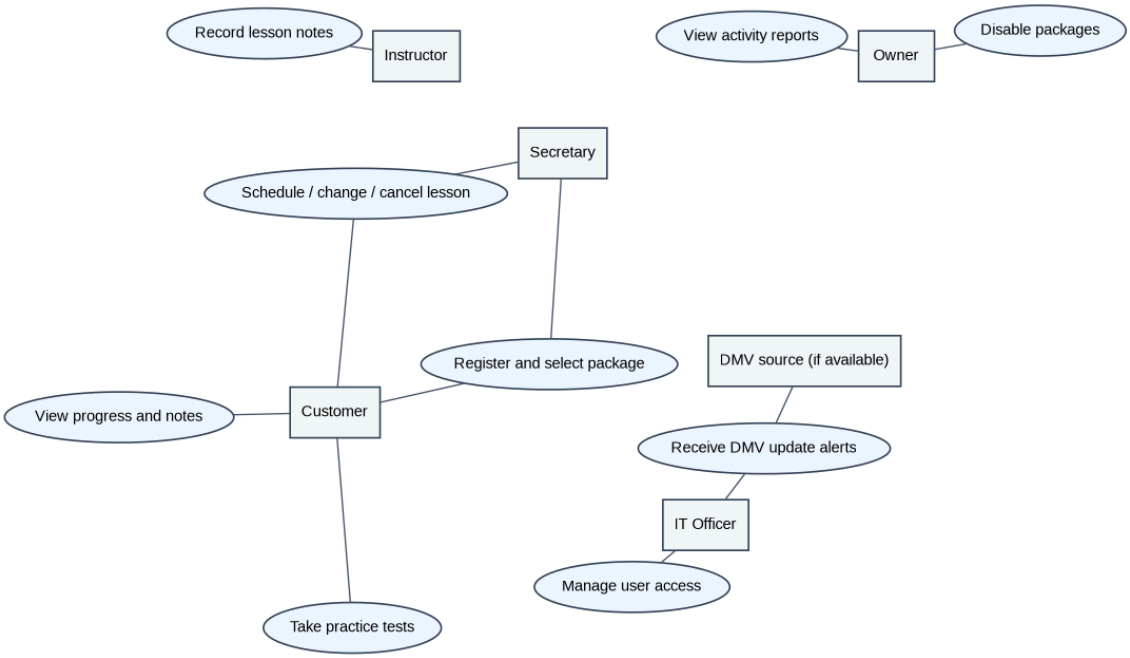
Southern New Hampshire University

CS 255 System Design Document Template

DriverPass | System Design Document | CS 255

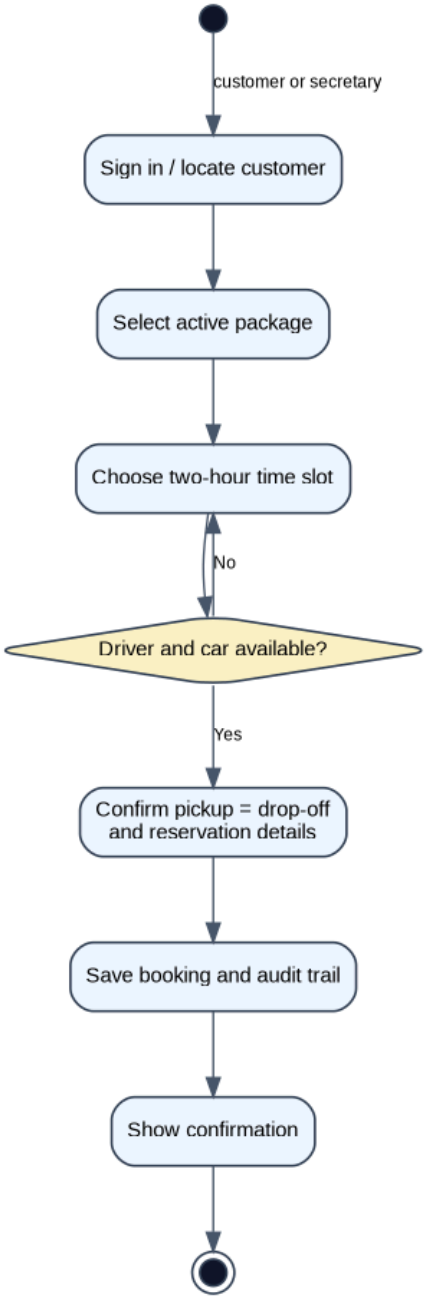
UML Diagrams

UML Use Case Diagram



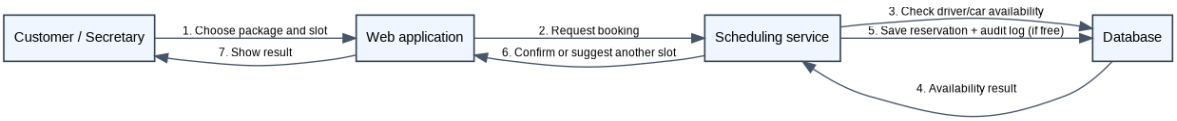
The diagram shows how customers, secretaries, instructors, management, and IT staff interact with DriverPass. The DMV update source is an external dependency, not a guaranteed integration.

UML Activity Diagrams



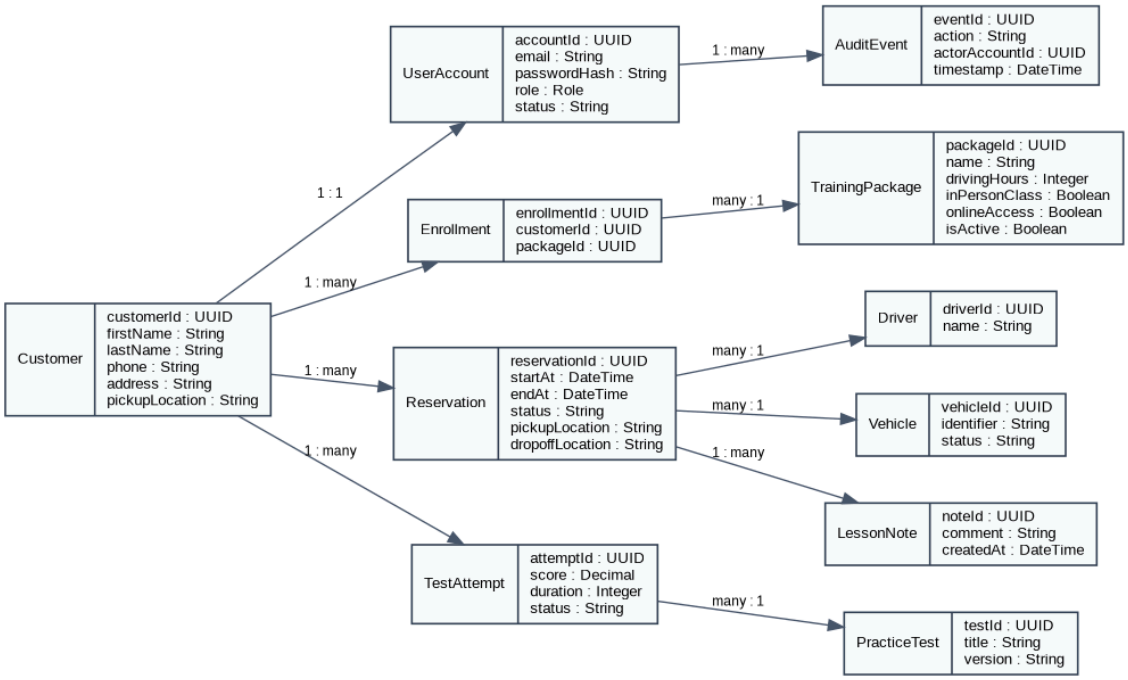
Activity 1: Scheduling a two-hour driving lesson. A booking is saved only after driver and vehicle availability are checked.

UML Sequence Diagram



Sequence for scheduling a lesson. The booking service checks availability before saving a reservation and an activity record.

UML Class Diagram



Proposed classes, attributes, and associations. Cardinalities are design assumptions where the interview does not specify implementation details.

Technical Requirements

Hardware: Customers and employees need a computer, tablet, or smartphone with a modern browser and internet connection. The cloud provider supplies server and storage hardware; DriverPass has ten cars, each paired with a driver.

Software: A responsive web application, an application service for scheduling and training, a relational database for customers, packages, reservations, test results and audit records, and a reporting/export feature compatible with Excel. Password-reset email delivery and a payment processor are proposed supporting services.

Infrastructure: Cloud hosting, encrypted HTTPS connections, managed database backups, access controls by job role, audit logging, and monitoring. Customers can download reports for offline analysis, but cannot modify live data offline.

Business rules: Lessons last two hours. Package One provides six driving hours (three lessons); Package Two provides eight driving hours (four lessons) and an in-person class; Package Three provides twelve driving hours (six lessons), an in-person class, and online lessons/practice tests. The system must prevent double-booking of drivers and vehicles and record who creates, changes, or cancels appointments. Staff can disable a package, but adding/removing modules is outside the initial release.

Security and data handling: Store password hashes rather than plain-text passwords; limit access to authorized roles; allow IT to disable/reset accounts; use payment-provider tokenization rather than retaining card security codes. Record changes with actor and time.

External dependency and assumptions: DMV updates and notifications depend on an available approved DMV feed or manual import process; the interview does not specify an API. Performance targets, uptime guarantees, hosting vendor, and payment vendor were not defined by the client and require confirmation.